

Econometrics II – Assignment 1:

Forecasting Economic Growth and Growth-at-Risk

Spring 2026

1 Introduction

GDP growth forecasts are key inputs to monetary, fiscal, and prudential policy, while the conditional lower tail—Growth-at-Risk (GaR)—is used by central banks to quantify recession risk (Adrian et al., 2019). We ask whether a parsimonious univariate model for U.S. quarterly GDP growth can deliver one-step-ahead point forecasts and 10% GaR estimates that outperform naive benchmarks over 2010Q1–2024Q4. We estimate AR(1)–AR(4) and ARMA(1,1) on 1990Q1–2009Q4 and evaluate 60 out-of-sample observations.

BIC and residual diagnostics favor AR(2): its forecasts are not statistically different from the naive historical-mean benchmark (Diebold-Mariano $p = 0.24$, HAC), and neither AR(2)-based nor unconditional GaR is rejected by the 10% coverage test, consistent with recent evidence that simple univariate models remain competitive (Adrian et al., 2025).

2 Description of the Data

The data consist of quarterly observations from 1990Q1 to 2024Q4. The variable of interest is $y_t = \log(\text{GDPC1}_t) - \log(\text{GDPC1}_{t-4})$, the log year-on-year growth rate of U.S. Real GDP (GDPC1), retrieved from the FRED database; the year-on-year transformation removes seasonal effects without seasonal adjustment.

Figure 1 plots the full sample. The series fluctuates around a positive mean with no apparent trend, consistent with stationarity. Two episodes stand out: the Global Financial Crisis (2008–2009) and the COVID-19 trough in 2020Q2 (roughly -9% , followed by a sharp V-shaped recovery); the AR order is determined from the ACF/PACF in §4.1.

Figure 2 shows the sample autocorrelation function (ACF) and partial autocorrelation function (PACF) for the training period 1990Q1–2009Q4. The ACF decays gradually

over several lags, indicating persistence, while the PACF drops sharply after lag 2 with remaining partial autocorrelations close to zero. This pattern is the classic signature of an AR(2) process, and guides our subsequent model selection.

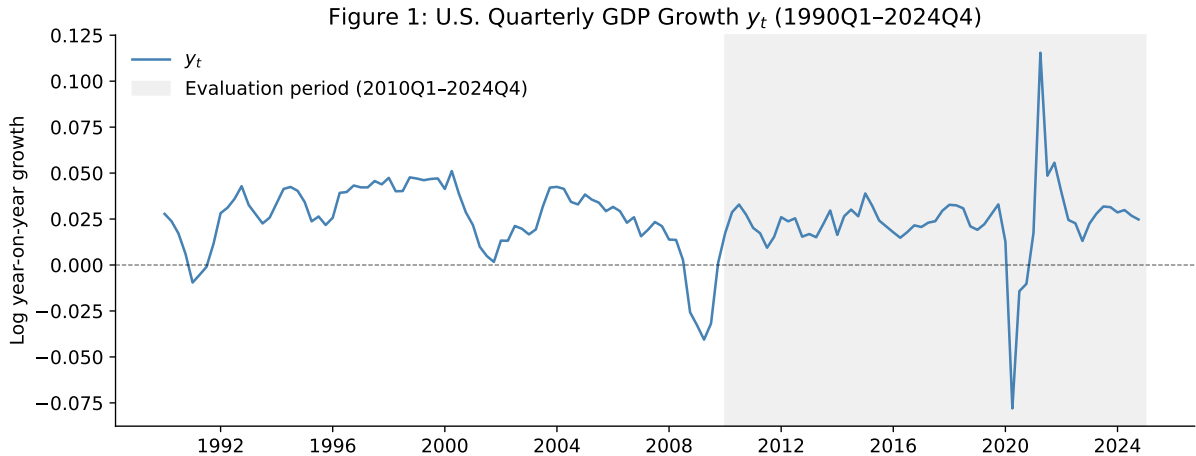


Figure 1: U.S. quarterly GDP growth y_t , 1990Q1–2024Q4. Shaded region indicates the out-of-sample evaluation period (2010Q1–2024Q4).

3 Econometric Theory

We consider the ARMA(p, q) model

$$y_t = \delta + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}, \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2), \quad (1)$$

where y_t is the log year-on-year GDP growth rate and δ is the intercept; a pure AR(p) corresponds to $q = 0$. Writing $\phi(L) = 1 - \phi_1 L - \cdots - \phi_p L^p$, the process is stationary and weakly dependent iff all roots z_j of $\phi(z) = 0$ satisfy $|z_j| > 1$ (equivalently, all inverse roots lie inside the unit circle). For $p = 2$ this reduces to the triangle $\phi_1 + \phi_2 < 1$, $\phi_2 - \phi_1 < 1$, $|\phi_2| < 1$. Under stationarity, y_t admits an MA(∞) representation with unconditional mean $\mu = \delta / (1 - \phi_1 - \cdots - \phi_p)$.

3.1 Estimation

For AR(p) models, parameters are estimated by OLS, which under $\varepsilon_t \sim N(0, \sigma^2)$ coincides with MLE. Under stationarity, weak dependence, and $E[\varepsilon_t | \mathcal{I}_{t-1}] = 0$, $\hat{\phi}$ is consistent and asymptotically normal, so a single coefficient is tested by the Wald t -ratio $H_0: \phi_j = 0$, $t_{\phi_j} = \hat{\phi}_j / \widehat{\text{s.e.}}(\hat{\phi}_j) \xrightarrow{d} N(0, 1)$.

For ARMA(p, q) with $q \geq 1$, OLS is infeasible (lagged ε_t are unobserved); we estimate by

Gaussian MLE with the conditional log-likelihood (given $y_1, \dots, y_p$)

$$\ell(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2) = -\frac{T}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^T \varepsilon_t^2,$$

where residuals are computed recursively.

Model selection follows a general-to-specific (GETS) strategy: starting from AR(4), nested restrictions are tested by the likelihood-ratio statistic

$$\xi_{\text{LR}} = -2(\log L(\tilde{\boldsymbol{\theta}}) - \log L(\hat{\boldsymbol{\theta}})) \xrightarrow{d} \chi^2(j)$$

for j restrictions. Because LR is sensitive to non-normality (and invalid under QMLE), each reduction is cross-checked with AIC and BIC; we use BIC as the primary criterion for its stronger complexity penalty.

3.2 Forecasting

The one-step-ahead forecast from (1) is $\hat{y}_{t+1} = E[y_{t+1} \mid \mathcal{I}_t]$, with parameters fixed at their training-sample estimates and residuals updated recursively via $\hat{\varepsilon}_t = y_t - \hat{y}_t$ for $t > T$. Forecasts are compared to the naive benchmark $\bar{y} = T^{-1} \sum_{t=1}^T y_t$ by RMSE and the Diebold–Mariano (DM) test on the squared-error differential $d_t = e_{\text{model},t}^2 - e_{\text{naive},t}^2$: under $H_0: E[d_t] = 0$, $t_{\text{DM}} = \bar{d} / \widehat{\text{s.e.}}(\bar{d}) \xrightarrow{d} N(0, 1)$, with $\widehat{\text{s.e.}}(\bar{d})$ a HAC estimator (autocorrelated d_t).

3.3 Growth-at-Risk

The one-period Growth-at-Risk at risk level $\alpha \in (0, 1)$ is the α -quantile of $y_t \mid \mathcal{I}_{t-1}$: $P(y_t \leq \text{GaR}_t^\alpha \mid \mathcal{I}_{t-1}) = \alpha$. Under $\varepsilon_t \sim N(0, \sigma^2)$,

$$\text{GaR}_t^\alpha = \hat{y}_t + \sigma \Phi^{-1}(\alpha), \tag{2}$$

where $\Phi^{-1}(\alpha)$ is the standard-normal α -quantile.

Evaluation uses the hit indicator $\text{hit}_t = \mathbf{1}\{y_t < \text{GaR}_t^\alpha\}$ and violation rate $\hat{p} = T^{-1} \sum_t \text{hit}_t$; under correct specification (hit_t) is i.i.d. Bernoulli(α), and $H_0: E[\text{hit}_t] = \alpha$ is tested by $t = (\hat{p} - \alpha) / \sqrt{\alpha(1 - \alpha)/T} \xrightarrow{d} N(0, 1)$. Magnitudes are captured by the tick loss $\mathcal{T}_\alpha(y_t, \text{GaR}_t^\alpha) = (\alpha - \text{hit}_t)(y_t - \text{GaR}_t^\alpha)$.

4 Empirical Analysis

4.1 Model Selection

Figure 2 motivated a focus on AR models: the PACF is statistically significant at lags 1 and 2 and negligible thereafter. Following the GETS principle, we estimate AR(1)–AR(4) and ARMA(1,1) and compare them in Table 1, then apply sequential LR tests starting from AR(4) to justify reducing the model.

Testing AR(4) vs. AR(3) ($H_0: \phi_4 = 0$) gives $\xi_{LR} = 0.08$ ($p = 0.78$): fail to reject, reduce. Testing AR(3) vs. AR(2) ($H_0: \phi_3 = 0$) gives $\xi_{LR} = 2.29$ ($p = 0.13$): again fail to reject, yielding AR(2) as the GETS outcome. Information criteria confirm: AR(2) achieves the best BIC (-542.33), while AR(1) is excluded due to significant residual autocorrelation ($Q(8)$, $p = 0.00$, see Table 1). We therefore select **AR(2)** as the preferred specification.

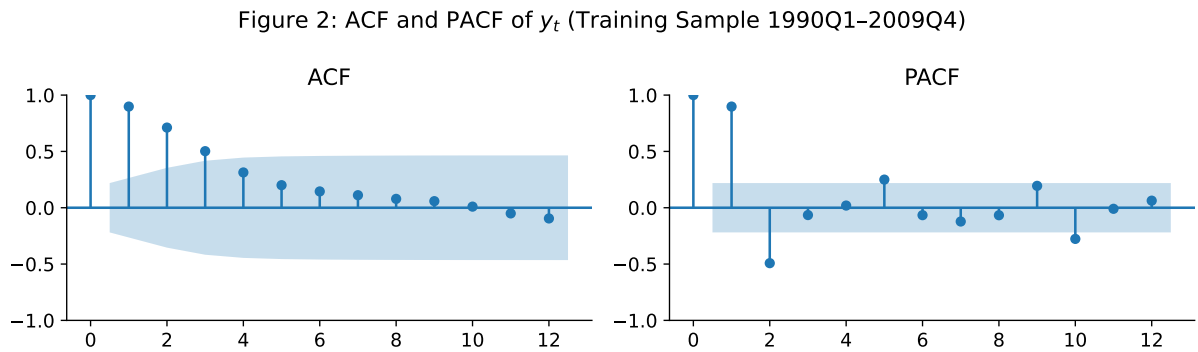


Figure 2: Sample ACF and PACF of y_t , training sample 1990Q1–2009Q4. Shaded regions indicate the $\pm 1.96/\sqrt{T}$ 95% confidence bands.

Table 1: Model comparison: AIC, BIC, $\hat{\sigma}$, and Ljung-Box $Q(8)$ p-value (training sample 1990Q1–2009Q4). Best value per column in bold.

Model	AIC	BIC	$\hat{\sigma}$	LB p-val (lag 8)
AR(1)	-535.78	-528.64	0.01	0.00
AR(2)	-551.86	-542.33	0.01	0.14
AR(3)	-552.15	-540.24	0.01	0.24
AR(4)	-550.23	-535.94	0.01	0.28
ARMA(1,1)	-546.13	-536.60	0.01	0.12

4.2 Misspecification Testing

Three misspecification tests are applied to the AR(2) training-sample residuals.

The Ljung-Box test (adjusted for $p = 2$ AR parameters, $\sim \chi^2(h - p)$) yields $Q(4) = 8.67$ ($p = 0.013$) and $Q(8) = 12.25$ ($p = 0.057$). The borderline rejections are attributable to the 2008Q4 GFC outlier inflating nearby autocorrelations, not to a failure of the AR(2) lag structure.

The Jarque-Bera test yields $JB = 8.63$ ($p = 0.013$), rejecting normality due to excess kurtosis of 1.31 from GFC residuals in 2008Q4–2009Q1; adding intervention dummies for these quarters reduces it to $JB = 0.26$ ($p = 0.879$), confirming the rejection is outlier-driven. We retain the plain AR(2), so the estimator is formally a QMLE: t -statistics remain valid under the sandwich covariance $\Omega^* = \mathcal{I}^{-1} \mathcal{J} \mathcal{I}^{-1}$, but the Gaussian GaR (2) may mis-calibrate the lower tail.

The ARCH-LM test at four lags yields 7.98 ($p = 0.092$), failing to reject conditional homoskedasticity at the 5% level.

4.3 Preferred Model: Estimates and Interpretation

Table 2 reports the ML estimates for the AR(2) model over 1990Q1–2009Q4. The estimated unconditional mean is $\hat{\mu} = 0.0262$ (standard error 0.006), corresponding to an average annual GDP growth of approximately 2.6%, which is in line with the U.S. long-run average. Both AR coefficients are highly significant: $\hat{\phi}_1 = 1.359$ ($t = 11.0$) and $\hat{\phi}_2 = -0.512$ ($t = -4.1$). All three AR(2) triangle conditions hold ($\hat{\phi}_1 + \hat{\phi}_2 = 0.847 < 1$, $\hat{\phi}_2 - \hat{\phi}_1 = -1.871 < 1$, $|\hat{\phi}_2| = 0.512 < 1$), so the process is stationary; the inverse roots are complex conjugates with modulus ≈ 0.72 , implying damped oscillations around the mean. The residual standard deviation $\hat{\sigma} = 0.010$ enters the GaR formula as the scale of the conditional distribution.

Table 2: Estimation results for the AR(2) model (training sample 1990Q1–2009Q4).

	Coefficient	Std. Error	t-stat	p-value
const	0.0262	0.0060	4.3455	0.0000
ar.L1	1.3587	0.1235	11.0055	0.0000
ar.L2	-0.5118	0.1241	-4.1237	0.0000
sigma2	0.0001	0.0000	6.5740	0.0000

4.4 Forecast Evaluation

Figure 3 and Table 3 present the out-of-sample results. The AR(2) RMSE is 0.0256 versus 0.0206 for the naive benchmark. The Diebold-Mariano statistic $t_{DM} = 1.186$ ($p = 0.236$,

HAC) fails to reject equal MSE; the gap is driven primarily by the COVID-19 shock, which lies outside the dynamics of any linear AR model.

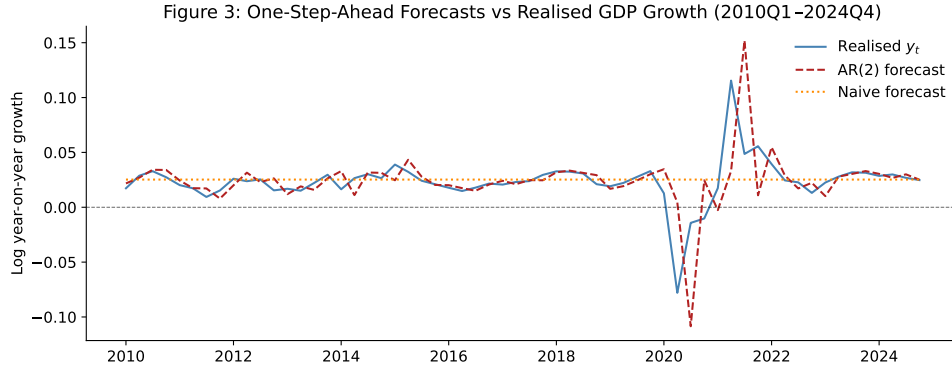


Figure 3: Out-of-sample forecasts vs realised GDP growth, 2010Q1–2024Q4.

Table 3: Forecast evaluation: RMSE and Diebold-Mariano test (2010Q1–2024Q4).

Model	RMSE	DM statistic	DM p-value
AR(2)	0.0256	1.1860	0.2360
Naive	0.0206	—	—

4.5 Growth-at-Risk Evaluation

Figure 4 and Table 4 present the GaR evaluation.

The AR(2) GaR has a violation rate of $\hat{p} = 0.150$; the coverage t -test ($t = 1.291$, $p = 0.197$) cannot reject correct coverage. The GaR^{np} series shows the mirror pattern ($\hat{p} = 0.050$, same $|t|$, $p = 0.197$), also not rejected. The AR(2) excess violations are largely attributable to the two COVID-19 quarters of 2020. On tick loss, GaR^{np} achieves a marginally lower mean loss (0.0040 vs. 0.0046). Mechanically, excess kurtosis (cf. JB in §4.2) pushes the true 10%-quantile below $\hat{y}_t + \hat{\sigma}\Phi^{-1}(0.10)$, so Gaussian GaR understates risk in turbulent quarters such as 2020. Overall, the AR(2) GaR provides useful time-variation relative to the constant benchmark but does not achieve demonstrably better calibration over this evaluation window.

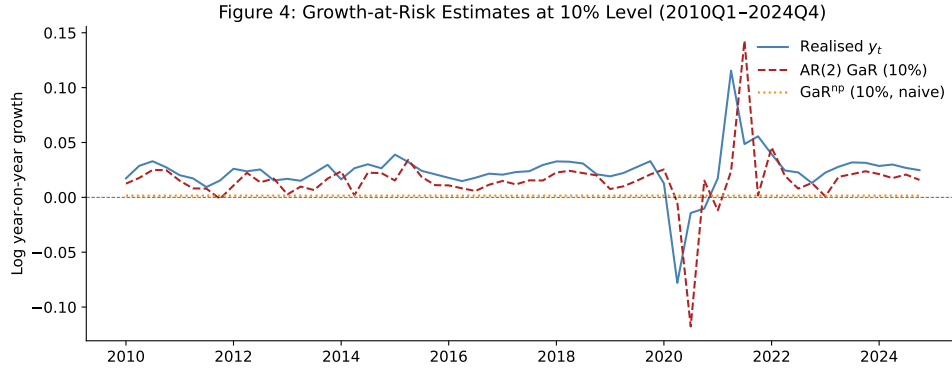


Figure 4: Growth-at-Risk estimates (10% level) vs realised GDP growth, 2010Q1–2024Q4.

Table 4: GaR evaluation at 10% level: violation rates, coverage test, and tick loss (2010Q1–2024Q4).

GaR series	Violation rate $\hat{p}$	t -stat ($H_0: \hat{p} = 0.10$)	p -value	Mean tick loss
AR(2) GaR	0.1500	1.2910	0.1970	0.0046
GaR ^{np} (naive)	0.0500	-1.2910	0.1970	0.0040

5 Discussion

Three limitations deserve mention. First, the Gaussian derivation of (2) is contradicted by the JB rejection (§4.2), so the non-rejection of the 10% coverage test should be read as weak evidence of calibration, not validation of the full conditional distribution.

Second, parameters are estimated once on 1990Q1–2009Q4 and held fixed, ruling out coefficient drift, level shifts, and changing shock volatility during the evaluation window; the univariate specification additionally ignores macro-financial state variables that may carry incremental information about downside risk.

Third, the evaluation window is short and dominated by rare extremes: the COVID-19 collapse and rebound drive a disproportionate share of squared errors and GaR exceedances, so the headline metrics are sensitive to a handful of quarters and the conclusions are best read as evidence of robustness through an unusually turbulent sample rather than a universal performance statement.

6 Conclusion

An AR(2) for U.S. quarterly GDP growth was selected by GETS and confirmed by BIC, with misspecification tests passing at the 5% level (JB rejection outlier-driven; t -statistics valid under QMLE). Out-of-sample, forecasts are indistinguishable from the naive benchmark (DM $p = 0.24$) and both GaR series pass the 10% coverage test; the dominant driver

of deviations is the COVID-19 shock, consistent with simple univariate models remaining competitive in macroeconomic forecasting and risk quantification.

References

- Adrian, T., N. Boyarchenko, and D. Giannone (2019). Vulnerable Growth. *American Economic Review*, 109(4), 1263–1289.
- Adrian, T., H. Chen, M.-S. Dovi, and J.H. Lee (2025). Machine-learning Growth-at-Risk. *arXiv:2506.00572*.